Differential Calculus

*Home work sheet #1

Limit

Find the limit of the following functions:

1. $\lim\limits_{x\to 0} \dfrac{x}{\sqrt{x+1}-1}$

2. $\lim\limits_{x\to 2} \dfrac{2x^2-5x+2}{5x^2-7x-6}$

3. $f(x)=\begin{cases} x^2+1 & ,x>0 \\ 1 & ,x=0 \\ 1+x & ,x<0 \end{cases}$

   Find $\lim\limits_{x\to 0} f(x)$

4. $f(x)=\begin{cases} 3x-1 & ,x<1 \\ 3-x & ,x>1 \end{cases}$

   Find $\lim\limits_{x\to 1} f(x)$

5. $\lim\limits_{x\to 0} \dfrac{x}{|x|}$

6. $\lim\limits_{x\to \infty} \dfrac{3x+5}{6x-8}$

7. $f(x)=\begin{cases} 2-x & ,x<1 \\ x^2+1 & ,x>1 \end{cases}$

   Find $\lim\limits_{x\to 1} f(x)$

8. $f(x)=\begin{cases} e^{\frac{-|x|}{2}} & ,-1<x<0 \\ x^2 & ,0<x<2 \end{cases}$

   Find $\lim\limits_{x\to 0} f(x)$

9. $f(x)=\begin{cases} \dfrac{1}{x+2} & ,x<-2 \\ x^2-5 & ,-2<x<3 \\ \sqrt{x+13} & ,x>3 \end{cases}$

   Find $\lim\limits_{x\to -2} f(x)$ and $\lim\limits_{x\to 3} f(x)$

10. $f(x)=\begin{cases} x^2 & ,x<1 \\ 2.4 & ,x=1 \\ x^2+1 & ,x>1 \end{cases}$

    Does $\lim\limits_{x\to 1} f(x)$ exist?

11. $\lim\limits_{x\to \infty} \left(\sqrt{x^6+5x^3}-x^3\right)$

12. Prove that $\lim\limits_{x\to \infty} \left(1+\dfrac{1}{x}\right)^x = e$

13. $\lim\limits_{x\to \infty} \sqrt[3]{\dfrac{3x+5}{6x-8}}$

14. $\lim\limits_{x\to -\infty} \dfrac{4x^2-x}{2x^3-5}$

15. Let $f(x)=\begin{cases} 2x+1 & ,x<1 \\ 3-x & ,x>1 \end{cases}$, find $\lim\limits_{x\to 1} f(x)$.

**These problems are for the students only as home work. Search the reference books for more examples.*

Differential Calculus

*Home work sheet # 2

Continuity and Differentiability

(a) Test the continuity of the following functions:

1. $f(x)=\begin{cases} \cos x, & x\ge 0 \\ -\cos x, & x<0 \end{cases}$ at $x=0$.

2. $f(x)=\begin{cases} x\cos(1/x), & x\ne 0 \\ 0 & ,x=0 \end{cases}$ at $x=0.$

3. $f(x)=\begin{cases} e^{1/x}, & x\ne 0 \\ 1 & ,x=0 \end{cases}$ at $x=0.$

4. $f(x)=\begin{cases} e^{\frac{-|x|}{2}} & ,-1<x<0 \\ x^2 & ,0\le x<2 \end{cases}$ at $x=0.$

5. $f(x)=\begin{cases} (x-a)\sin\left(\dfrac{1}{x-a}\right) & ,x\ne a \\ 0 & ,x=a \end{cases}$ at $x=a.$

6. $f(x)=\begin{cases} 1 & ,x<0 \\ 1+\sin x & ,0\le x<\pi/2 \\ 2+(x-\pi/2)^2 & ,x\ge \pi/2 \end{cases}$
at $x=0$ and $x=\pi/2.$

7. $f(x)=\begin{cases} x\sin(1/x) & ,x\ne 0 \\ 0 & ,x=0 \end{cases}$ at $x=0.$

8. $f(x)=\begin{cases} \dfrac{|x-3|}{x-3} & ,x\ne 3 \\ 0 & ,x=3 \end{cases}$ at $x=3.$

9. $f(x)=|x|+|x-1|$ at $x=0$ and $x=1.$

10. $f(x)=\begin{cases} (1+x)^{1/x} & ,x\ne 0 \\ 1 & ,x=0 \end{cases}$ at $x=0.$

(b) Test the differentiability of the following functions:

1. $f(x)=\begin{cases} x^2\cos(1/x) & ,x\ne 0 \\ 0 & ,x=0 \end{cases}$ at $x=0.$

2. $f(x)=\begin{cases} x^2\sin(1/x) & ,x\ne 0 \\ 0 & ,x=0 \end{cases}$ at $x=0.$

3. $f(x)=|x|$ at $x=0.$

(c) Let $f(x)=\begin{cases} x^2-16x & ,x<9 \\ 12\sqrt{x} & ,x\ge 9 \end{cases}.$

Is $f(x)$ continuous at $x=9$? Determine whether $f(x)$ is differentiable at $x=9$.

(d) Let $f(x)=\begin{cases} x^2 & ,x\le 1 \\ \sqrt{x} & ,x>1 \end{cases}.$

Is $f(x)$ continuous at $x=1$? Determine whether $f(x)$ is differentiable at $x=1.$

(e) Show that $f(x)=\begin{cases} x^2+1 & ,x\le 1 \\ x & ,x>1 \end{cases}$ is neither continuous nor differentiable at $x=1.$ Also sketch the graph of $f(x)$.

These problems are for the students only as home work. Search the reference books for more examples.

Differential Calculus

*Home work sheet # 3

Techniques of Differentiation

1. Find the differential coefficients of the following functions with respect to *x*. (i.e. $\frac{dy}{dx}$).

$$(i)\, y=\sin x \sin 2x \sin 3x,\ \ (ii)\, y=\operatorname{cosec}^3 x,\ \ (iii)\, y=\cos 2x\cos 3x,\ \ (iv)\, y=\sin^{-1}(x^2),$$

$$(v)\, y=\tan(\sin^{-1}x),\quad (vi)\cot^{-1}\left(\frac{1+x}{1-x}\right),\quad (vii)\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right),\quad (viii)\sin^{-1}\left(\frac{2x}{1+x^2}\right),$$

$$(ix)\tan^{-1}\left(\frac{2x}{1-x^2}\right),\quad (x)\tan^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right),\quad (xi)\sin\left(2\tan^{-1}\sqrt{\frac{1-x}{1+x}}\right),\quad (xii)\ln\sqrt{\frac{1-\cos x}{1+\cos x}}.$$

2. Find the differential coefficients of:

$$(i)(\sin x)^{\ln x},\ \ (ii)(\sin x)^{\cos x}+(\cos x)^{\sin x}.$$

3. Find $\frac{dy}{dx}$ in the following cases:

$$(i)\, 3x^4-x^2y+2y^3=0,\quad (ii)\, x^3+y^3+4x^2y-25=0,\quad (iii)\, x^y=y^x.$$

4. Find $\frac{dy}{dx}$ when

$$(i)\, x=a\cos^3\theta,\ y=a\sin^3\theta,\quad (ii)\, x=\sin^2\theta, y=\tan\theta,\quad (iii)\, x=a\sec^2\theta, y=a\tan^2\theta.$$

5. Differentiate the left-side functions with respect to the right-side ones:

$$(i)\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)\text{with respect to }\tan^{-1}\left(\frac{2x}{1-x^2}\right)\quad (ii)\, x^{\sin^{-1}(x)}\text{ with respect to }\sin^{-1}x.$$

**These problems are for the students only as home work. Search the reference books for more examples.*

Differential Calculus

*Home work sheet # 4

Maxima and minima

1. Find (a) the open intervals on which f is increasing , (b) the open intervals on which f is decreasing, (c) the open intervals on which f is concave up, (d)) the open intervals on which f is concave down and (e) the x- coordinate of all inflection points.

   $(i)\ f(x) = x^2 - 5x + 6 \qquad (ii)\ f(x) = 5 + 12x - x^3$

   $(iii)\ f(x) = x^4 - 8x^2 + 16 \qquad (iv)\ f(x) = \dfrac{x^2}{x^2+2} \qquad (v)\ f(x) = \sqrt[3]{x+2}.$

2. Locate the critical numbers and identify which critical numbers correspond to stationary points.

   $(i)\ f(x) = x^3 + 3x^2 - 9x + 1 \quad (ii)\ f(x) = x^4 - 6x^2 - 3$

   $(iii)\ f(x) = \dfrac{x}{x^2+2} \qquad (iv)\ f(x) = x^{2/3}$

   $(v)\ f(x) = x^{1/3}(x+4) \qquad (vi)\ f(x) = \cos 3x.$

3. Find the relative extrema (maxima/ minima) using both the first and second derivative tests.

   $(i)\ f(x) = 2x^3 - 9x^2 + 12x \quad (ii)\ f(x) = \dfrac{x}{2} - \sin x, \quad 0 < x < 2\pi.$

4. Use the given derivative to find all critical numbers of $f(x)$ and determine whether a relative maximum, relative minimum, or neither occurs there.

   $(i)\ f'(x) = x^3(x^2 - 5) \qquad (ii)\ f'(x) = \dfrac{x^2-1}{x^2+1}.$

**These problems are for the students only as home work. Search the reference books for more examples.*

Differential Calculus

*Home work sheet # 5

<u>Successive Differentiation</u>

(a) Find the n-th derivative of the following functions:

1. $y = x^n$
2. $y = (ax+b)^n$
3. $y = \ln(ax+b)$
4. $y = \dfrac{1}{x+a}$
5. $y = e^{ax}$
6. $y = \sin(ax+b)$
7. $y = \cos(ax+b)$

(b). If $y = e^{ax}\sin bx,$ then show that $y_2 - 2ay_1 + (a^2+b^2)y = 0.$

(c). If $y = e^{x}\sin x,$ then show that $y_4 + 4y = 0.$

<u>Rolle's and Mean Value Theorem</u>

(a). Verify the hypothesis of Rolle's Theorem for the following functions:

1. $f(x) = x^2 - 6x + 8; \quad [2,4]$
2. $f(x) = \cos x; \quad [\pi/2, 3\pi/2]$
3. $f(x) = \dfrac{x}{2} - \sqrt{x}; \quad [0, 4].$

(b). Verify the hypothesis of Mean Value Theorem for the following functions:

1. $f(x) = x^3 + x - 4; \quad [-1,2]$
2. $f(x) = \sqrt{x+1}; \quad [0,3]$
3. $f(x) = \sqrt{25-x^2}; \quad [0,5].$

<u>Maclaurin and Taylor Series</u>

(a). Find the Taylor series for the following functions:

(i) $\sin x,$ at $x_0 = \dfrac{\pi}{2}.$ (ii) $\ln x,$ at $x_0 = 2.$

(b). Expand $y = \ln x$ in the power of $x-2$ and $y = e^{ax}$ in the power of $x-1$.

(c). Find the Maclaurin series for the function e^{ax} and $\cos x$.

(d). Find the Maclaurin polynomial p_0, p_1, p_2, p_3 for $e^x \cos x$.

(e). Expand $y = \ln(x+1)$, $y = \sin x$, $y = \cos x$ in the power of x.

**These problems are for the students only as home work. Search the reference books for more examples.*

Differential Calculus

*Home work sheet # 6

Leibnitz's Theorem

1. If $y=\tan^{-1}x$, then show that $(1+x^2)y_{n+2}+2(n+1)xy_{n+1}+n(n+1)y_n=0.$
2. If $y=\cot^{-1}x,$ then show that $(1+x^2)y_{n+2}+2(n+1)xy_{n+1}+n(n+1)y_n=0.$
3. If $y\sqrt{1-x^2}=\sin^{-1}x$, then show that $(1-x^2)y_{n+1}-(2n+1)xy_n-n^2y_{n-1}=0.$
4. If $y=e^{\tan^{-1}x}$, then show that $(1+x^2)y_{n+2}+(2nx+2x-1)y_{n+1}+n(n+1)y_n=0.$
5. If $y=e^{m\sin^{-1}x}$, then show that $(1-x^2)y_{n+2}-(2n+1)xy_{n+1}-(n^2+m^2)y_n=0.$
6. If $y=(\sin^{-1}x)^2$, then show that $(1-x^2)y_{n+2}-(2n+1)xy_{n+1}-n^2y_n=0.$
7. If $\log_e y=a\sin^{-1}x$ then show that $(1-x^2)y_{n+2}-(2n+1)xy_{n+1}-(n^2+a^2)y_n=0.$
8. If $y=e^{m\cos^{-1}x}$ then show that $(1-x^2)y_{n+2}-(2n+1)xy_{n+1}-(n^2+m^2)y_n=0.$
9. If $\log_e y=\tan^{-1}x$ then show that $(1+x^2)y_{n+2}+(2nx+2x-1)y_{n+1}+n(n+1)y_n=0.$
10. If $y=\left(\cos^{-1}x\right)^2$, then show that $(1-x^2)y_{n+2}-(2n+1)xy_{n+1}-n^2y_n=0.$
11. If $\ln y=m\cos^{-1}x,$ then show that $(1-x^2)y_{n+2}-(2n+1)xy_{n+1}-(n^2+m^2)y_n=0.$
12. If $x=\tan(\ln y),$ then show that $(1+x^2)y_{n+2}+(2nx+2x-1)y_{n+1}+n(n+1)y_n=0.$

Indeterminate Forms

Find the limit using L' Hospital's rule:

$$1.\lim_{x\to1}\frac{\ln x}{x-1},\quad 2.\lim_{x\to3}\frac{x-3}{3x^2-13x+12},\quad 3.\lim_{x\to0}\left(\frac{1}{x^2}-\frac{\cos 3x}{x^2}\right),\quad 4.\lim_{x\to\pi}\frac{\sin x}{x-\pi}$$

$$5.\lim_{x\to0}\frac{x-\tan^{-1}x}{x^3},\quad 6.\lim_{x\to+\infty}\frac{e^{3x}}{x^2},\quad 7.\lim_{x\to0}\frac{a^x-1-x\log a}{x^2},\quad 8.\lim_{x\to0}(e^x+x)^{\frac{1}{x}}$$

$$9.\lim_{x\to0}\left(\frac{1}{x}-\frac{1}{\sin x}\right),\quad 10.\lim_{x\to\pi}(x-\pi)\cot x,\quad 11.\lim_{x\to0}\frac{\ln(\sin x)}{\ln(\tan x)},\quad 12.\lim_{x\to\infty}xe^{-x}$$

$$13.\lim_{x\to0}\frac{\sin 2x}{x},\quad 14.\lim_{x\to0}\frac{\sin x}{x^2},\quad 15.\lim_{x\to\infty}\frac{x}{e^x},\quad 16.\lim_{x\to0}\left(\frac{1}{x}-\frac{1}{xe^x}\right),\quad 17.\lim_{x\to0}\frac{\sin 2x}{\sin 5x}.$$

**These problems are for the students only as home work. Search the reference books for more examples.*